

Week 4 — Session 4

Topic: Matrix Basics

Language: C++17

Focus: matrix traversal, diagonal processing, and transformation through transposition.

Task 1: Diagonal Difference

Practice: <https://www.hackerrank.com/challenges/diagonal-difference/>

Calculate the absolute difference between the sums of the two diagonals of a square matrix.

Solution Code

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    int n;
    cin >> n;

    long long primary = 0, secondary = 0;

    for (int i = 0; i < n; ++i) {
        for (int j = 0; j < n; ++j) {
            long long x;
            cin >> x;

            if (i == j)
                primary += x;

            if (i + j == n - 1)
                secondary += x;
        }
    }

    cout << llabs(primary - secondary) << '\n';
    return 0;
}
```

Task 2: Transpose Matrix

Practice: <https://leetcode.com/problems/transpose-matrix/>

Create the transpose of a matrix by swapping its row and column positions.

Solution Code

```
#include <bits/stdc++.h>
using namespace std;

vector<vector<int>> transpose(const vector<vector<int>>& matrix) {
    int m = matrix.size();
    int n = matrix[0].size();

    vector<vector<int>> result(n, vector<int>(m));

    for (int i = 0; i < m; ++i)
        for (int j = 0; j < n; ++j)
            result[j][i] = matrix[i][j];

    return result;
}

int main() {
    int m, n;
    cin >> m >> n;

    vector<vector<int>> matrix(m, vector<int>(n));

    for (auto& row : matrix)
        for (int &x : row)
            cin >> x;

    auto result = transpose(matrix);

    for (const auto& row : result) {
        for (int x : row)
            cout << x << ' ';
        cout << '\n';
    }
}
```

```
    }  
    return 0;  
}
```

All programs are standalone C++17 solutions and can be compiled with a standard C++17 compiler.